

MAT 310, Handout 25: Method of Moment-Generating Functions (6.5)

Method of MGFs: We can find the distribution of $U = f(Y_1, Y_2, \dots, Y_n)$ by calculating its mgf, using knowledge of the mgfs of $Y_1, Y_2, \dots, Y_n$, and comparing it to well-known mgfs.

Theorem. Let $m_X(t)$ = mgf of RV X and $m_Y(t)$ = mgf of RV Y . If $m_X(t) = m_Y(t)$, then X and Y have the same distribution. In other words, the mgf uniquely defines the distribution.

Example 1. Let $Y \sim U(0, 1)$. Use the method of mgfs to find the distribution of $X = 4 + 3Y$. Recall: for $W \sim U(a, b)$, we know $m_W(t) = \frac{e^{tb} - e^{ta}}{t(b - a)}$.

Recall: $Y \sim N(\mu, \sigma)$ has mgf $m_Y(t) = e^{\mu t + \frac{\sigma^2 t^2}{2}}$.

Example 2. If $m_X(t) = e^{-2t+4t^2}$, what is the distribution of X ?

Example 3. If $m_W(t) = e^{t^2}$, what is the distribution of W ?

Theorem. If $Y_1, Y_2, \dots, Y_n$ are independent RVs with mgfs $m_1(t), m_2(t), \dots, m_n(t)$ and $X = Y_1 + Y_2 + \dots + Y_n$, then:

$$m_X(t) =$$

Proof:

Theorem. If $Y_1, Y_2, \dots, Y_n$ are independent Normal RVs s.t. $Y_i \sim N(\mu_i, \sigma_i)$ and $X = \sum_{i=1}^n a_i Y_i$, then:

$$X \sim$$

Proof:

Example 4. If $X \sim N(-2, 3)$ and $Y \sim N(3, 2)$ are independent, find the distribution of:

(a) $W_1 = X - 2Y$

(b) $W_2 = 3X + 4Y$